

MATH 465: COMBINATORICS AND GRAPH THEORY SYLLABUS

Fall 2026

1. VITAL INFORMATION

Instructor: Yelena Mandelshtam
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Section 002 class times and locations: TuTh 2:30pm–4:00pm (EH 1068)
Section 001 class times and locations: TuTh 4:00pm–5:30pm (EH 1084)
Office Hours: Tu 1:15pm–2:15pm, W 1:15pm–2:15pm, F 1:45pm–2:45pm,
and by appointment. (Hours subject to change.)

Please come and talk to me in office hours!

1.1. Important Dates

Classes Start	Monday, August 31
First Quiz	Thursday, September 10
Last day to add a course	Monday, September 21
Last day to drop a course	Monday, September 21
Exam 1 (in-class)	Thursday, October 15th
Fall break	October 19-20
Last day to late drop a course	Friday, November 6
Thanksgiving break	November 25-27
Exam 2 (in-class)	Tuesday, December 8th
Last day of classes	Friday, December 11

1.2. Course Description and Objectives

This is an introductory combinatorics class at the undergraduate level. Combinatorics is the study of finite mathematical objects, including their enumeration, structural properties, design, and optimization. Combinatorics plays an increasingly important role in various branches of mathematics and in numerous applications, including computer science, statistics and statistical physics, operations research, bioinformatics, and electrical engineering. The first half of the course will cover the fundamentals of enumerative combinatorics. The second half will cover the fundamentals of graph theory.

1.3. Prerequisites

The most important prerequisite is mathematical maturity. You have to be comfortable with reading and writing proofs. In particular, proofs by contradiction and proofs by induction will be common and used without further explanation. Some experience with proof-based linear algebra is assumed.

1.4. Textbook

There is no required textbook for this course.

1.5. Canvas

Class resources will be available through Canvas.

2. ASSESSMENTS

2.1. Homework

Practicing is important! Homework sets will be assigned regularly and submitted via Gradescope. Some subset of the problems will be graded for accuracy. Late work will not be accepted for a grade, but the lowest homework grade will be dropped. Homework solutions may be handwritten or typed with $\text{\LaTeX}$, but must be neat and legible. You may discuss problems with other students in the class, but you must write up your solutions independently in your own words and turn in your own work. Please also indicate who you worked with at the top of your submission.

2.2. Weekly quizzes

There will be a short 10–15 minute quiz in class each Thursday. Each quiz will either consist of a randomly selected question from the previous week's homework or a simple question testing your understanding of a definition or other concept from class. These quizzes are not intended to be difficult; their purpose is to encourage you to keep up with the course material and make sure that you are comfortable with the basic concepts as we go. Your two lowest quiz grades will be dropped. This is intended to account for occasional illness, travel, emergencies, or other unavoidable absences. In general, there will be no make-up quizzes. If you expect to miss multiple quizzes because of an extended illness, disability accommodation, or other exceptional circumstance, please talk to me so we can make appropriate arrangements.

2.3. Exams

There will be two in-class 80-minute exams. Each exam will cover one half of the course. These exams **will not** be cumulative.

- **Exam 1 (Enumerative combinatorics):** Thursday, October 15, in class.
- **Exam 2 (Graph theory and related topics):** Tuesday, December 8, in class.

Note, a majority of exam questions will be based closely on homework assignments.

3. COURSE GRADE

Your course grade will be determined as follows:

Written Homework	20 points
Quizzes	20 points
Exam 1	30 points
Exam 2	30 points
100 points total	

The number of points you earn will be mapped to a letter grade as follows:

A+: [98, 100]	A: [93, 98)	A-: [90, 93)	B+: [87, 90)	B: [83, 87)	B-: [80, 83)
C+: [77, 80)	C: [73, 77)	C-: [70, 73)	D+: [67, 70)	D: [63, 67)	D-: [60, 63)

Depending on scores, I may adjust to a more generous letter grade assignment. The class will NOT be graded on a curve, and there is no set distribution of final grades I intend to follow.

4. POLICIES

4.1. Attendance and Classroom Etiquette

Regular attendance is expected as is full engagement in classwork activities. Please arrive on time, put away your cell phone, and stay for the entire class period. **You may not use cell phones during class.** If you use other devices for note-taking, please use them exclusively for that purpose during class time. Studies suggest that student multi-tasking during class through use of smart phones and laptops hinders classroom learning for both users and *nearby peers*.

If you are sick, please do not attend class. If you must have an extended absence or repeated absences, please email me so we can make alternative arrangements.

4.2. Learning Needs

The University of Michigan is committed to ensuring the full participation of all students, and I am committed to making learning as accessible as possible for all of my students. If you have a disability and need an accommodation to participate in this class or to complete course requirements, please ask the Services for Students with Disabilities (SSD) office (G664 Haven Hall; <https://studentaccessibility.umich.edu/ssd>) to provide documentation of the accommodations that you need. Then, SSD can provide me with documentation so we can make sure you have the resources you need. I will treat any information that you share as private and confidential. If you have a letter from SSD, **please come talk to me as soon as possible** so we can figure out what you need in order to participate fully in the course and make appropriate arrangements for quizzes, exams, and other course requirements.

4.3. Academic Integrity

If you cheat, you will fail the assessment and may fail the course. Sometimes first year students are unaware of academic norms and cheat unintentionally. Some examples can be found on the LSA Community Standards of Academic Integrity <https://lsa.umich.edu/lsa/academics/academic-integrity.html>. If you have questions about what is or is not allowed in this course, please ask.

4.4. Generative AI

The capabilities of generative AI tools continue to rapidly evolve and change. At present, freely available tools are able to accurately solve (or almost solve) pretty much every homework problem in an introductory math class, and much more. However, homework is meant primarily for you to practice problem-solving and writing skills that you will need, at minimum, on the exams. Thus, you should not and are not permitted to use such tools (or other external resources) to provide solutions to homework problems, nor to “proofread” submitted solutions. However, you may use such tools to clarify and test your understanding of concepts and examples from lectures, but be aware that generative AI tools can and do make mistakes. If you are unclear about whether or not a specific tool or use case is permitted, please talk to me. It is important that we maintain an open dialogue as we all continue to learn best practices around learning with generative AI tools.

4.5. Grading Concerns / Regrade requests

After receiving a graded exam, quiz, or homework, you have 1 week (7 days) to raise concerns about grading errors.

5. RESOURCES AND TIPS FOR SUCCESS

5.1. Tips for success

- Use class time wisely: fully engage yourself in class activities, asking and answering questions when appropriate.
- Seek understanding rather than trying to rely on memorized facts or formulas.
- It can be tempting to ask an LLM all of your questions, and this can be useful in many cases. However, mathematics is a human endeavor, and it is important to talk to people! Ask questions in class, come to office hours, and discuss mathematics with your classmates.

5.2. Office Hours

Office hours are a time for students to drop by the instructor's office without an appointment to ask questions or discuss matters related to the course. However, it is expected that you will have already formulated your questions and attempted to answer them prior to showing up at office hours. Note that office hours are not remedial and, in fact, many of the strongest students regularly attend office hours. If you think you would like to come to office hours at any point in the semester, and none of the listed times work for you, please talk to me as soon as possible!

5.3. Mental health and well-being

As a student, you may experience challenges that negatively affect your learning, such as anxiety, depression, interpersonal or sexual violence, difficulty with eating or sleeping, grief/loss, and alcohol or drug problems. The University of Michigan offers several confidential services that you might find helpful for addressing such challenges, including

- Counseling and Psychological Services (CAPS): 734-764-8312
- Psychiatric Emergency Services: 734-996-4747
- Sexual Assault Prevention and Awareness Center (SAPAC) 24-Hour Crisis Line: 734-936-3333
- If you have a diagnosed mental health condition, you may also be able to register with Services for Students with Disabilities: 734-763-3000